

Last time (X, d) metric space, $A \subseteq X$.

① A is disconnected $\Leftrightarrow \exists B, C \subseteq X$ s.t.:

- $A = B \cup C$
- $B \neq \emptyset, C \neq \emptyset$
- $\overline{B} \cap C = \emptyset, B \cap \overline{C} = \emptyset$

② A is disconnected in $X \Leftrightarrow A$ is disconnected in A (with the induced metric)

- So, it suffices to prove facts about the whole space being connected, rather than just a subset

Prop Let (X, d) be a metric space. Then:

X is connected $\Leftrightarrow$ the only subsets of X that are both open and closed are $\emptyset$ and X .

Pf: $\Rightarrow$: Suppose not: X is connected, but $\exists$ a set $A \neq \emptyset, X$ that is both open and closed.

Set $B = A^c$. Then:

① $A \cup B = A \cup A^c = X$

② $A \neq \emptyset, B = A^c \neq \emptyset$ (since $A \neq X$)

③ A and B are separated:

$$\begin{aligned} \overline{A} \cap B &= A \cap A^c = \emptyset, & A \cap \overline{B} &= A \cap \overline{A^c} = \emptyset \\ \underbrace{\overline{A}}_{= A} & \text{ (since } A \text{ is closed)} & \underbrace{\overline{A^c}}_{= A^c} & \text{ (since } A \text{ open} \\ & & & \Rightarrow A^c \text{ closed)} \end{aligned}$$

So X is disconnected — a contradiction.

⇐: Suppose not: the only subsets of X that are both open and closed are $\emptyset$ and X , but X is disconnected. Then $\exists A, B \subseteq X$ s.t.:

- ① $X = A \cup B \Rightarrow B = A^c$
- ② $A \neq \emptyset, B \neq \emptyset \Rightarrow A \neq \emptyset, X$
- ③ A and B are separated:

$$\emptyset = \bar{A} \cap B = \bar{A} \cap A^c$$

$$\Rightarrow \bar{A} = A$$

$$\Rightarrow A \text{ is closed}$$

$$\emptyset = A \cap \bar{B} = A \cap \overline{A^c}$$

$$\Rightarrow \overline{A^c} = A^c$$

$$\Rightarrow A^c \text{ closed} \Rightarrow A \text{ open}$$

So $A \neq \emptyset, X$ is a set that is both closed and open — a contradiction. $\square$

Prop Let (X, d) be a metric space and let $\emptyset \neq C \subseteq X$ be connected. If $A, B \subseteq X$ are separated sets and $C \subseteq A \cup B$, then either $C \subseteq A$ or $C \subseteq B$.

Pf: Set $\tilde{A} = A \cap C$ and $\tilde{B} = B \cap C$. Then:

$$\textcircled{1} \tilde{A} \cup \tilde{B} = (A \cap C) \cup (B \cap C) = (A \cup B) \cap C = C$$

② $\tilde{A}$ and $\tilde{B}$ are separated:

$$\tilde{A} \subseteq A, \quad \tilde{B} \subseteq B \Rightarrow \begin{cases} \bar{\tilde{A}} \cap \tilde{B} \subseteq \bar{A} \cap B = \emptyset \\ \tilde{A} \cap \bar{\tilde{B}} \subseteq A \cap \bar{B} = \emptyset \end{cases}$$

As C is connected, then we cannot have both $\tilde{A} \neq \emptyset$ and $\tilde{B} \neq \emptyset$. So either:

$$\tilde{A} = \emptyset \Rightarrow C = \tilde{A} \cup \tilde{B} = \tilde{B} = B \cap C \Rightarrow C \subseteq B$$

or

$$\tilde{B} = \emptyset \Rightarrow C = \tilde{A} \cup \tilde{B} = \tilde{A} = A \cap C \Rightarrow C \subseteq A. \quad \square$$

• Next: special case $X = \mathbb{R}$

Def An interval is a subset of $\mathbb{R}$ of the form

(a, b)	(a, ∞)	$\mathbb{R}$
$[a, b]$	$[a, \infty)$	$\emptyset$
$[a, b)$	$(-\infty, b]$	$\{a\}$
$[a, b]$	$(-\infty, b]$	

for some $a, b \in \mathbb{R}$.

Thm If $I \subseteq \mathbb{R}$ is an interval, then I is connected in $(\mathbb{R}, |\cdot|)$.

Pf: Fix an interval $I \subseteq \mathbb{R}$. Note that

$$x, z \in I, \quad x < y < z \Rightarrow y \in I$$

by transitivity. E.g., for $I = (a, b)$,

$$a < x < y < z < b \Rightarrow a < y < b$$

• The other 10 cases are similar.

Suppose towards a contradiction that I were disconnected. Then $\exists A, B \subseteq \mathbb{R}$ s.t.

① $I = A \cup B$

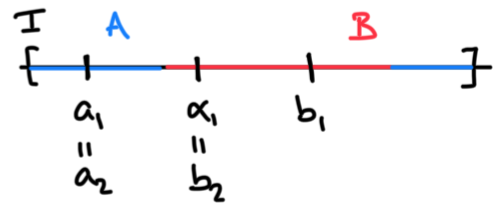
② $A \neq \emptyset, B \neq \emptyset$

③ A and B are separated.

As $A, B \neq \emptyset$, $\exists a_1 \in A$ and $b_1 \in B$. We may assume $a_1 < b_1$, after swapping A and B if necessary.

Set $\alpha_1 = \frac{a_1 + b_1}{2}$. Then

$a_1 < \alpha_1 < b_1$, so $\alpha_1 \in I = A \cup B$.



① Case $\alpha_1 \in A$: Set $(a_2, b_2) = (\alpha_1, b_1)$.

② Case $\alpha_1 \in B$: Set $(a_2, b_2) = (a_1, \alpha_1)$.

In both cases, $a_1 \leq a_2 \leq b_2 \leq b_1$, $a_2 \in A$, $b_2 \in B$.

Set $\alpha_2 = \frac{a_2 + b_2}{2}$. Then $\alpha_2 \in I = A \cup B$.

① Case $\alpha_2 \in A$: Set $(a_3, b_3) = (a_2, b_2)$.

② Case $\alpha_2 \in B$: Set $(a_3, b_3) = (a_2, \alpha_2)$.

Then $a_2 \leq a_3 \leq b_3 \leq b_2$, $a_3 \in A$, and $b_3 \in B$.

Proceeding inductively, we get:

① an increasing sequence $\{a_n\}$ in A , bounded above by b_1

② a decreasing sequence $\{b_n\}$ in B , bounded below by a_1

So $a = \lim a_n$ and $b = \lim b_n$ exist in $\mathbb{R}$. Note that $a_1 \leq a \leq b_1$ and $a_1 \leq b \leq b_1$, so $a, b \in I = A \cup B$.

Moreover,

$$a = \lim a_n \in \bar{A}, \quad \bar{A} \cap B = \emptyset \Rightarrow a \notin B \Rightarrow a \in A$$

$$b = \lim b_n \in \bar{B}, \quad A \cap \bar{B} = \emptyset \Rightarrow b \notin A \Rightarrow b \in B$$

- Recall: $x \in \bar{A} \Leftrightarrow \exists$ a sequence in A that converges to x

On the other hand,

$$b_{n+1} - a_{n+1} = \frac{b_n - a_n}{2} = \frac{b_{n-1} - a_{n-1}}{4} = \dots = \frac{b_1 - a_1}{2^n} \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\Rightarrow a = b \quad \Rightarrow A \cap B \neq \emptyset$$

This contradicts that A and B were separated. $\square$